Statistical Modelling & Analysis (STAT2004)
Applied Class (Week 2)

We will go through Question 2, 5 & 4 (if time permits) together in the Applied Class. Try the other questions yourself during the week and discuss your results with your tutor next week.

1. Suppose $X_1, \ldots, X_n$ is a random sample from an $\mathsf{Bernoulli}(p)$ distribution. Laplace's estimator of the proportion p is
$$\hat{p}_L = \frac{(\sum_{j=1}^n X_j) + 1}{n+2}.$$
   (a) What is the sampling distribution of $S = \sum_{j=1}^n X_j$
   (b) Is $\hat{p}_L$ an unbiased estimator of p? If not, determine the bias of $\hat{p}_L$.
   (c) Determine the mean-squared error of $\hat{p}_L$.
   (d) For which values of p does $\hat{p}_L$ have smaller MSE than the usual estimator $\hat{p} = n^{-1}\sum_{j=1}^n X_j$.

2. Suppose $X_1, \ldots, X_n$ is a random sample from an $\mathsf{Exponential}(\lambda)$ distribution. Here $\lambda > 0$ is the rate parameter so the pdf is $f(x;\lambda) = \lambda\exp(-\lambda x)$ for $x \geq 0$ and $f(x;\lambda) = 0$ for $x < 0$. It may be helpful to know the following: If $Y \sim \mathsf{Gamma}(\alpha, \beta)$, then
$$\mathbb{E}(Y) = \frac{\alpha}{\beta}, \quad \mathrm{Var}(Y) = \frac{\alpha}{\beta^2}, \quad \mathbb{E}(Y^{-1}) = \frac{\beta}{\alpha - 1}, \quad \mathrm{Var}(Y^{-1}) = \frac{\beta^2}{(\alpha-1)^2(\alpha-2)}$$
and the moment generating function of Y is
$$M_Y(t) = \left(\frac{\beta}{\beta - t}\right)^{\alpha}, \text{ for } t < \beta.$$
The $\mathsf{Exponential}(\lambda)$ distribution coincides with the $\mathsf{Gamma}(1, \lambda)$ distribution.
   (a) Show the sampling distribution of $\bar{X}$ is $\mathsf{Gamma}(n, n\lambda)$ $(\star)$.
   (b) Determine the method of moments estimator for λ.
   (c) Determine the bias of the estimator from part (b).
   (d) Determine the constant c such that $c/\bar{X}$ is an unbiased estimator of λ.
   (e) Does the estimator from part (d) have a smaller MSE than the method of moments estimator? Justify your answer.

3. Suppose $X_1, \ldots, X_{2n+1}$ is a random sample. Our statistical model is $\mathcal{F} = \{f(x;\theta) = g(x-\theta); \theta \in \mathbb{R}\}$, where g is some fixed pdf that is symmetric about 0 and satisfying
$$\int_{-\infty}^{\infty} |x| g(x) dx < \infty.$$
Using results from ordered statistics (STAT2003 - Claim 46) it can be shown that the sample median M has sampling distribution with pdf
$$f_M(x;\theta) = \frac{(2n+1)!}{n!\,n!}[F(x;\theta)]^n[1 - F(x;\theta)]^n f(x;\theta), \qquad x \in \mathbb{R},$$
where $F(x;\theta)$ is the cdf for $f(x;\theta)$.
   (a) Show that $F(\theta + x;\theta) = 1 - F(\theta - x;\theta)$, for $x \in \mathbb{R}$.
   (b) Show that M is an unbiased estimator of θ.

4. Suppose $X_1, \ldots, X_n$ is a random sample from a $\mathsf{Geometric}(p)$ distribution, whose support is the positive integers $\{1, 2, \ldots\}$. As $\mathbb{E}_p(X) = 1/p$, the method of moments estimator is $\hat{p} = 1/\bar{X}$.

   (a) Is $\hat{p}$ a consistent estimator of p? Justify your answer.

   (b) Carry out a small simulation study to investigate the finite sample properties of $\hat{p}$. Choose a value of p in $(0, 1)$. For each sample size $\{10, 50, 100, 500, 1000\}$, generate $B = 10,000$ independent samples from the $\mathsf{Geometric}(p)$ distribution, and for each of these samples compute $\hat{p}$. For each value of n, use the results of the simulation to estimate the bias, variance, and MSE of $\hat{p}$. Scale these quantities by n, (ie calculate $n\times$bias, $n\times$variance, and $n\times$MSE) and present the results in the table. What do you notice?
   Note: The `R` function `rgeom` simulates from the geometric distribution supported on $\{0, 1, 2, \ldots\}$. To generate from the geometric distribution on $\{1, 2, \ldots\}$ you will need to add one to the simulated values.

5. The negative binomial distribution is used to model over-dispersed counts, that is counts where the variance is larger than the mean. We write $\mathsf{NB}(r, p)$ for the distribution with probability mass function

$$f(x) = \binom{x + r - 1}{r - 1} p^r (1 - p)^x, \quad x = 0, 1, 2, \ldots,$$

   where $r \in (0, \infty)$ and $p \in (0, 1]$. If $X \sim \mathsf{NB}(r, p)$, then

$$\mathbb{E}(X) = \frac{r(1-p)}{p} \quad \text{and} \quad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}.$$

   Let $X_1, \ldots, X_n$ be a random sample from $\mathsf{NB}(r, p)$.

   (a) Determine the method of moments estimator for (r, p).

   (b) Under what conditions does your method of moments estimator belong to the parameter space.